

Machine Learning

1. Setting the scene: definitions & objectives

Louis Bagot
Université Claude Bernard Lyon 1

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Course info

Websites & where to find info

- All administrative information on the [course website](#)
- Slides and labs for my part on the github, [Machine Learning](#) and [Deep Learning](#) parts
- All slides come with a “presentation mode” (the one I show in class) and a “reading mode” (with slides with notes) to read at home

Course sources

Course written by Louis Bagot.

Inspired by the following courses:

- Remy Cazabet, Sorbonne Université & Univ Lyon 1 (last year's course)
- Probabilistic Machine Learning, Kevin P. Murphy
- Understanding Deep Learning, by Simon J.D. Prince
- Machine Learning in Python with scikit-learn, by INRIA

See the slides with notes for references of individual slides (figures and notions) .

What is Machine Learning?

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Vague definition (Tom Mitchell):

A Computer Program is said to learn from **Experience E** with respect to some class of **Tasks T**, and **Performance** measure **P**, **if its Performance at Tasks** in **T**, as measured by **P**, **improves with Experience E**.

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→ What are E , T , P , and the Computer Program ??

What are the Experience E, Task T, Performance P?

→ Experience E =

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 - it's *because* the concept of data is vague/general that ML is powerful
- Loosely, we want to extract *knowledge* from the data (defined with T, P)

Experience E = DATA, but what is it?

Experience **E** = DATA, but what is it?

Generally comes as a

$$\text{Dataset } \mathcal{D} = \{\mathbf{x}_i\}_{i=1}^N$$

- each $\mathbf{x}_i$ is a **sample** (e.g., a person, image, sound, company, ...)

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- each $\mathbf{x}_i$ is a **sample** (e.g., a person, image, sound, company, ...)
- each sample is a vector with several **features**: $\mathbf{x}_i = \{x_i^j\}_{j=1}^d$
the features describe the sample (e.g. height, weight; colors, price, ...)

Experience E = DATA, what does it look like?

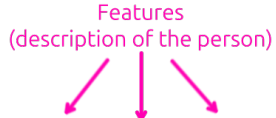
Experience E = DATA, what does it look like?

Dataset as a **matrix**:

(here, with the weight &
height of people)

8388 samples (people)
and each 3 features

Features
(description of the person)



	Weight (kg)	Standing Height (cm)	BMI(kg/m**2)
0	97.1	160.2	37.8
1	98.8	182.3	29.7
2	74.3	184.2	21.9
3	103.7	185.3	30.2
4	83.3	177.1	26.6
...	...	...	...
8383	94.3	178.8	29.5
8384	82.8	147.8	37.9
8385	108.8	168.7	38.2
8386	79.5	176.4	25.5
8387	59.7	167.5	21.3

8388 rows × 3 columns

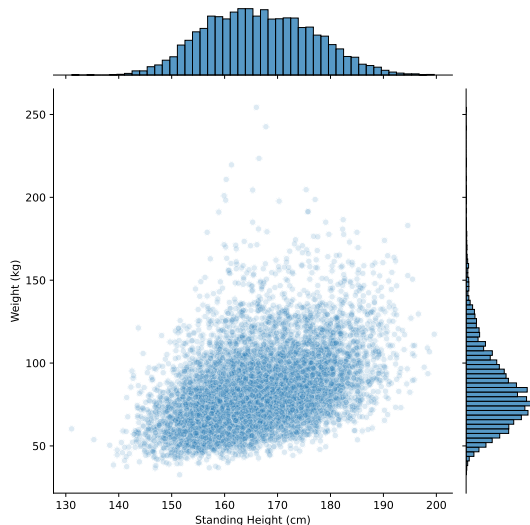
First sample
(person)



Experience **E** = DATA, what does it look like?

Think of the dataset as N
samples of an unknown
distribution over features:

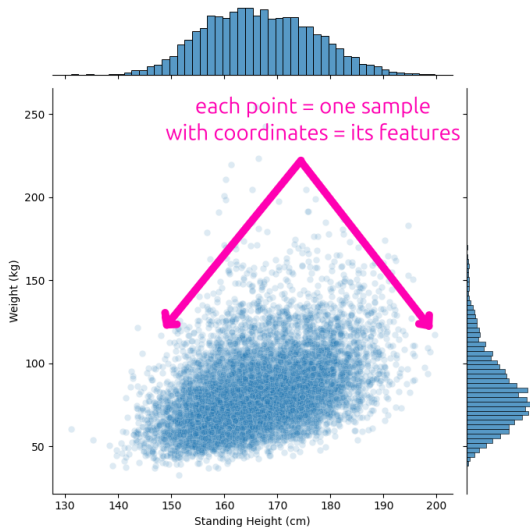
(each feature = one dimension)



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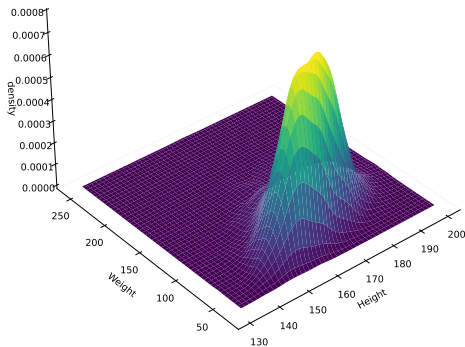
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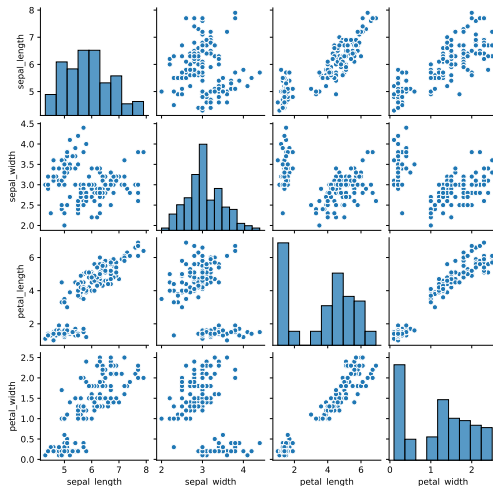
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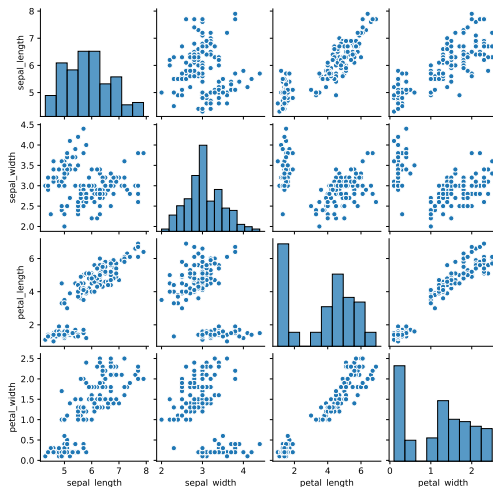
With more features (dimensions),
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Experience E = DATA, what does it look like?

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here, the dataset consists of 150
flowers described with 4 features:
petal/sepal height/width.



What are the **Experience E**, **Task T** and **Performance P**?

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→ **What is the Task T?**

What are we even trying to do with all that data?

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Machine Learning:

Match a desired function, $f \approx g : \mathcal{X} \rightarrow \mathcal{Y}$

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→ in ML we steal this name 😁

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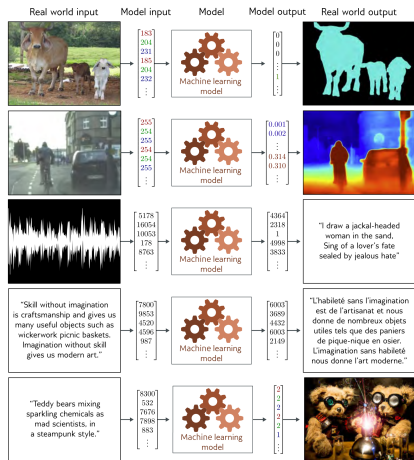
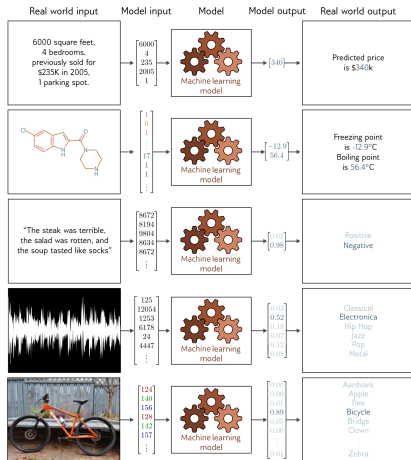
→ in ML we steal this name 😊

Task : the **Computer Program** tries to find a **model** f which approximates g ,

$$\underbrace{f}_{\text{model}} \left(\underbrace{\mathbf{x}_i}_{\text{input}} \right) = \underbrace{\hat{\mathbf{y}}_i}_{\substack{\text{output /} \\ \text{prediction}}} \approx g(\mathbf{x}_i)$$

Examples of functions g to approximate, please?

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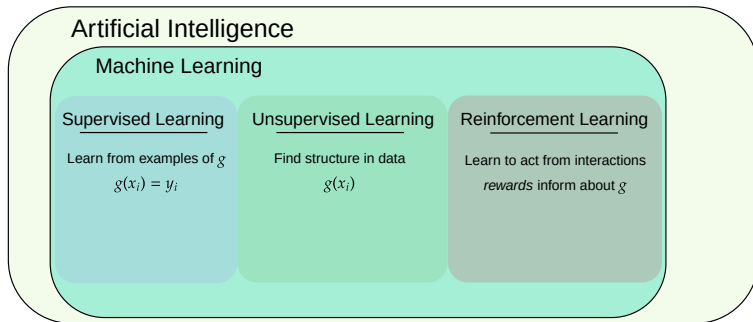
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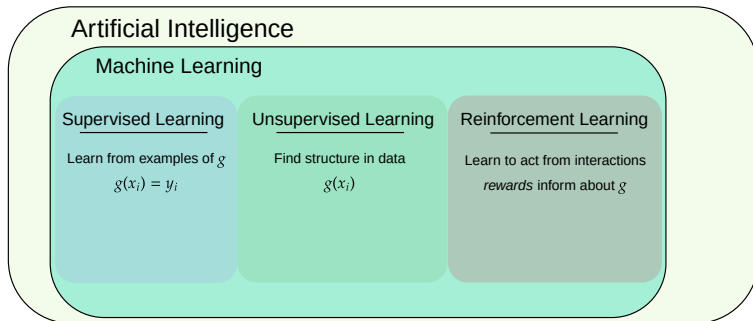
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Main focus of this course: **Supervised Learning**. the others use SL indirectly anyway :D

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- Can be gathered from natural processes (historic observations e.g. climate?)

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- Generally provided by humans (captcha-like problems)
- Can be gathered from natural processes (historic observations e.g. climate?)
- Major headache of Supervised Learning! It's costly!

Supervised Learning: what do the labels look like?

Supervised Learning: what do the labels look like?

Two main types of Supervised Learning:

- **Regression:** y_i are real values, e.g. temperature, prices...
 - predict the price of an apartment from size, location, number of rooms...
 - predict the number of passengers on all metro lines
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 - item on an image
 - grandmaster's move in a chess game

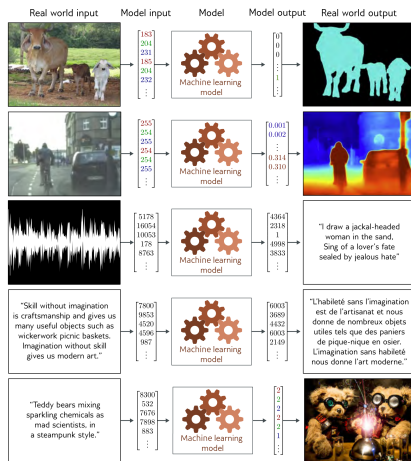
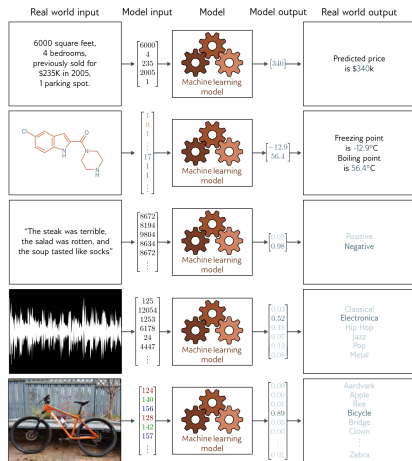
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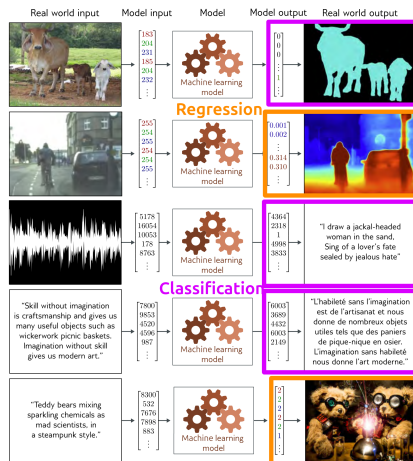
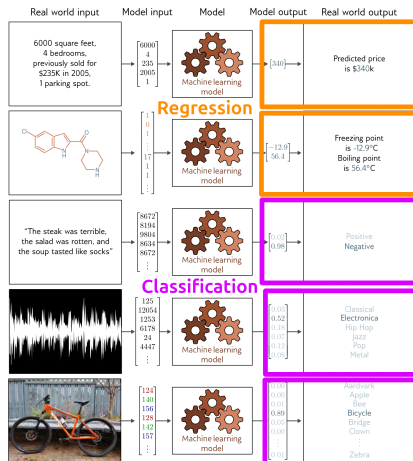
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In all cases our **Task** is to **approximate** g

Classification vs Regression: can you guess?

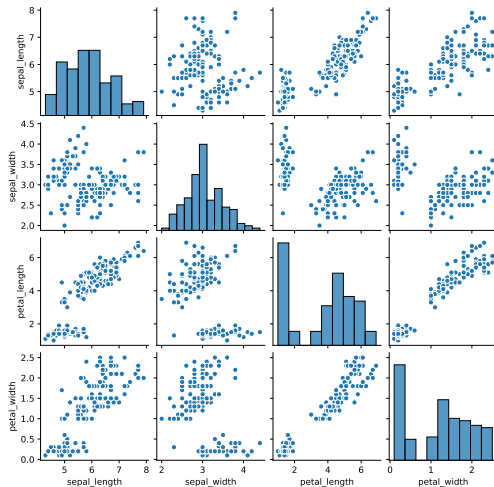


Classification vs Regression: can you guess?



What does it look like now?

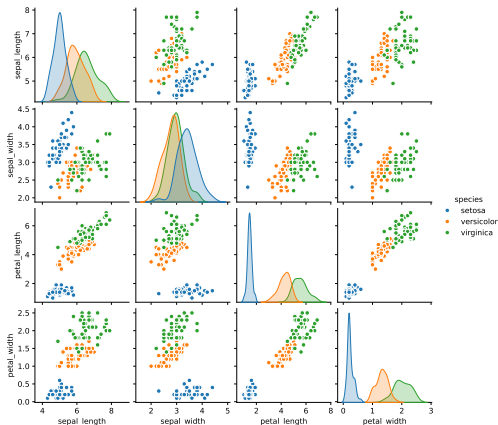
Still think of it as a distribution, but
some of the columns / dimensions
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What does it look like now?

Still think of it as a distribution, but
some of the columns / dimensions
are hidden from the model:

the flowers belong to 3 different species!



Supervised Learning: approximate g with the model f from examples:

$$f(\mathbf{x}_i) \approx \mathbf{y}_i = g(\mathbf{x}_i).$$

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→ **What is the Performance metric P?**

General idea:

Minimize distance between $f(\mathbf{x}_i)$ and $\mathbf{y}_i$

What is the **Performance** metric **P**?

→ what is the notion of distance between $f(x_i)$ and y_i ?

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To maximize the Performance, we minimize the Loss

Summary time!

What is Machine Learning ?

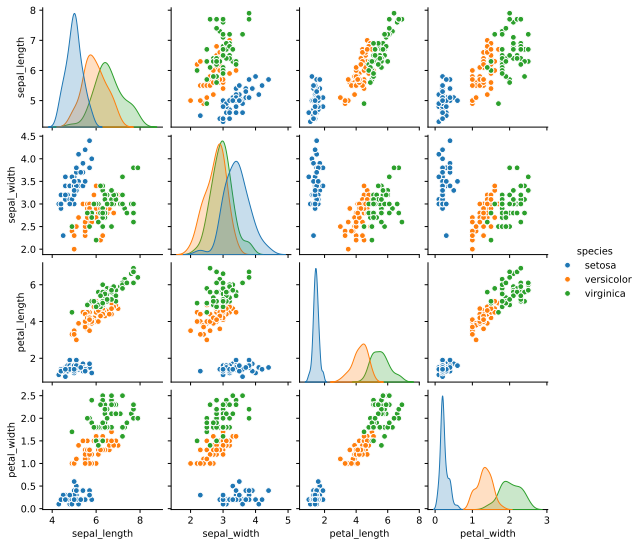
Summary time!

What is Machine Learning ?

The **Computer Program** learns a model to **predict the labels** from the **Data**,
as measured by the **decrease in prediction error** it makes

The Computer Program learns a model... How? What is it?

How would **you** tackle it?



Just pick the label of the closest thing?

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Yes!

Algo called **K-Nearest Neighbors** (KNN):

- pick a number of neighbors $k \geq 1$
- pick a notion of distance $\delta(\mathbf{x}_i, \mathbf{x}_j)$

Just pick the label of the closest thing?

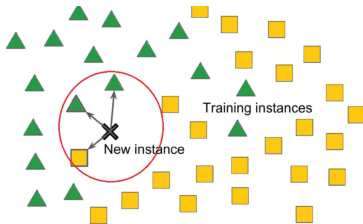
Yes!

Also called **K-Nearest Neighbors** (KNN):

- pick a number of neighbors $k \geq 1$
- pick a notion of distance $\delta(\mathbf{x}_i, \mathbf{x}_j)$

During inference for $\mathbf{x}$:

- compute the KNN
 $\arg \min_{\mathbf{x}_i \in \mathcal{D}} \delta(\mathbf{x}_i, \mathbf{x})$
- (classif) pick the most frequent class
- (regression) pick the average value



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K-Nearest Neighbors analysis:

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K-Nearest Neighbors analysis:

- simple, can be surprisingly good
- explainable → clear strengths and weaknesses
- ultra-reliant on a good distance
- needs to pass through the entire dataset at inference!
- ... did we learn anything? good baseline though!

How do we avoid passing through the whole dataset?

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Parametrized models:

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Parametrized models: math. functions with two sets of inputs, $f_{\theta}(\mathbf{x})$

- Usual inputs $\mathbf{x}$, e.g. a dataset sample, $\mathbf{x} = (x_1, x_2, \dots, x_d)$, each $x_i \in \mathbb{R}$
- The **parameters** θ , a set of scalars, $\theta = (\theta_1, \theta_2, \dots, \theta_M)$, each $\theta_i \in \mathbb{R}$
 - (sometimes called **weights**)

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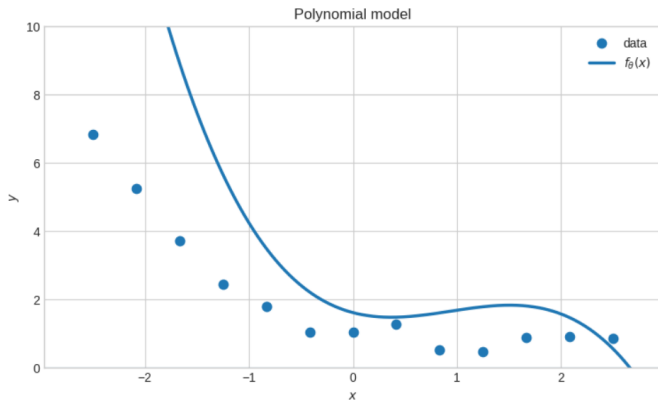
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 - (sometimes called **weights**)
- The parametrized model is just a big math. formula involving $\mathbf{x}$ and θ .

How do we avoid passing through the whole dataset?

Parametrized models, simplest example: **a linear or polynomial model**

$$f_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$$

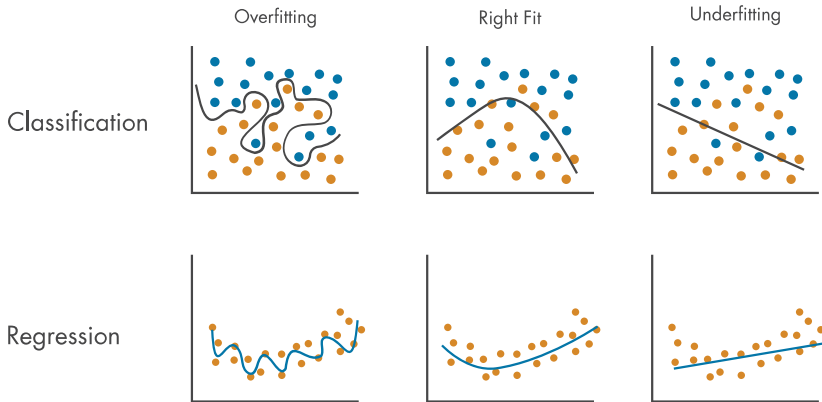


Parameters $\theta = [1.60, -0.80, 1.35, -0.48]$
Mean squared error $L(\theta) = 21.2285$

Why stop at degree 3? Let's make this model incredible!

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Problem: we *can* make models better like that... **but is that what we want?**



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Yes. Definition mistake: just minimizing the loss isn’t what we want

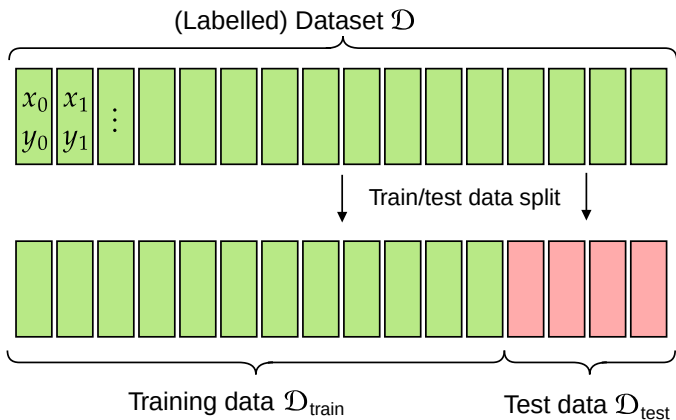
- the model can simply **memorize** the dataset examples!
- happens with too little data, too many parameters.

It’s in the **Task** and **Performance** metrics:

→ minimize the loss **on never-seen samples!**

How do we get samples we've never seen?

Implementation trick: split the dataset into test/train! (e.g. 30-70%)

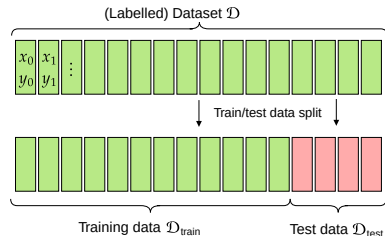


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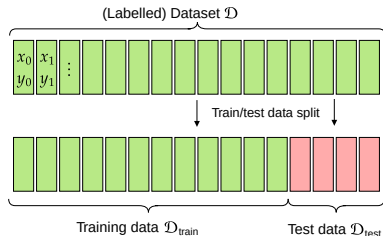


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You know you're overfitting when the **training performance is much better than the testing one!**

Important note: scientific integrity

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Machine Learning will feel like cooking a lot of the time.

- picking models
- picking hyperparameters
- applying random tricks & hacks

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- it informs about the real outcome you can expect from the model.

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Model evaluation is where the real science happens.

- train/test is like the placebo or A/B test for ML
- it informs about the real outcome you can expect from the model.
- easy to mess up: **never re-use the test data to fine-tune models**
 - otherwise it's like picking the perfect treatment for 10 specific people
 - re-sample new test data before re-training
 - use a *validation set* for hyperparameter tuning

Where were we?

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Think of all the possible curves you can generate with your parametrized model, by changing the parameters $\theta = (\theta_1, \theta_2, \dots)$.

Where were we?

Improve from KNN → **Parametrized models** $f_{\theta}(\mathbf{x})$

What are we trying to do with those?

Think of all the possible curves you can generate with your parametrized model, by changing the parameters $\theta = (\theta_1, \theta_2, \dots)$.

We are looking for **the curves that best fits the data**

→ the set of parameters θ with the **lowest loss**:

$$\arg \min_{\theta} \mathcal{L}(f_{\theta}, \mathcal{D})$$

($\mathcal{D}_{\text{train}}$ for training, $\mathcal{D}_{\text{eval}}$ for eval)

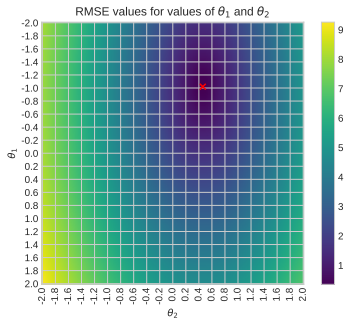
What does that look like?

New change of perspective:

Visualize the **weights as dimensions**,
(all values each weight can take in $\mathbb{R}$),
the **loss as a surface / color gradient**.

→ We are looking for the lowest spot!

$$\arg \min_{\theta} \mathcal{L}(f_{\theta}, \mathcal{D})$$



[\[click\] cool interactive version](#)

$$\arg \min_{\theta} \mathcal{L}(f_{\theta}, \mathcal{D})$$

This is called an **optimization problem**

→ we're looking for the *optimal* parameters, θ^*

How do we find the optimal parameters?

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How do we find the optimal parameters?

Many approaches possible: (we will see some, also see optim course)

- exact formula
- random shooting
- genetic algorithms
- gradient-based methods
- ...

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For an input with many dimensions, weight each input with some scalar weight:

$$\begin{aligned}f_{\theta}(\mathbf{x}) &= \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots \\ &= \mathbf{x}^{\top} \boldsymbol{\theta}\end{aligned}$$

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The simplest case of the polynomial!

For an input with many dimensions, weight each input with some scalar weight:

$$\begin{aligned}f_{\boldsymbol{\theta}}(\mathbf{x}) &= \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots \\&= \mathbf{x}^{\top} \boldsymbol{\theta}\end{aligned}$$

Writing the dataset design matrix X and dataset label vector $\mathbf{y}$, we can compute exactly the optimal (problem and solution called **Ordinary Least Squares**)

$$\boldsymbol{\theta}^{\star} \doteq \arg \min_{\boldsymbol{\theta}} \frac{1}{N} \sum_i (\mathbf{x}_i^{\top} \boldsymbol{\theta} - y_i)^2 = (X^{\top} X)^{-1} X^{\top} \mathbf{y}$$

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 - in general, this encodes “a bit more x consistently gives a bit more/less y ”
- it informs about linear relationships:
 - if θ_j is 0, then feature x_j is linearly irrelevant
 - if θ_j is positive, y generally increases with x_j
 - if θ_j is negative, y generally decreases with x_j
 - gives **insight!!** now we're learning!

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- this explains why **the linear model is still extremely popular**

Other losses/evaluation metrics

We have seen **Mean Squared Error**:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N (f_{\theta}(\mathbf{x}_i) - \mathbf{y}_i)^2$$

- most popular loss
- sensitive to outliers (very heavy, squared weight)
- interpretation issue: big value, hard to read...
→ can use Root MSE $RMSE = \sqrt{\text{MSE}}$

Other losses/evaluation metrics

Use the R^2 **score** for readability:

$$R^2 = 1 - \frac{\text{MSE}}{\text{Variance}} = 1 - \frac{\sum_{i=1}^N (f_{\theta}(\mathbf{x}_i) - \mathbf{y}_i)^2}{\sum_{i=1}^N (\bar{\mathbf{y}} - \mathbf{y}_i)^2}$$

- compares our prediction to the naive model “prediction = mean”
- perfect model: $\text{MSE}=0, R^2 = 1$
- naive model: $\text{MSE}=\text{Variance}, R^2 = 0$
- terrible model: $R^2 < 0$

Other losses/evaluation metrics

Mean Absolute Error:

$$\mathcal{L}_{\text{MAE}} = \frac{1}{N} \sum_{i=1}^N |f_{\theta}(\mathbf{x}_i) - \mathbf{y}_i|$$

- easier to read than MSE (in the unit of y)
- less sensitive to outliers
- less punitive of errors